

Second-harmonic minority ICRF heating in ITER-EDA

The nominal $N=2$ reference case

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Abstract

The reference second-harmonic ITER-EDA case is a mass-3 minority at 50% concentration in deuterium, heated at 60.9 MHz with the rigorous non-linear self-collision operator. It reaches a steady state in 12.5 s of physical time, at an effective temperature of 32.59 keV against a 12 keV bulk, with 73.9% of the tail energy perpendicular. The absorbed power density is 1.82 MW/m³, shed to the background ions and to the electrons in the ratio 59:41. The RF fills the tail on 0.22 s, faster than either collisional channel drains it, which is what sustains the supra-thermal population.

1 Configuration

A tritium-like minority (mass 3, $Z=1$) in a deuterium majority, both at 50% of $n_e = 1.5 \times 10^{20} \text{ m}^{-3}$, with $T_e = T_i = 12 \text{ keV}$ and $B_0 = 6 \text{ T}$. The wave is applied at the second cyclotron harmonic (Table 1). Self-collisions use the non-linear operator built from the Rosenbluth potentials of the evolving distribution (`isc` = -1); the time-differencing weight is $\theta = 0.75$ (`icn` = 1).

Table 1: Case parameters. The run terminated on the `conv_diag` criteria at $t = 12.5 \text{ s}$ after 276 s of wall time.

harmonic	$N = 2$	grid	150×150 , quadratic in $v_\perp$
frequency	60.9 MHz	$v_\perp^{\text{max}}$	$3.0 \times 10^7 \text{ m/s}$
$k_\perp, k_\parallel$	40, 2 m^{-1}	$v_\parallel$ range	$[-8, 8] \times 10^6 \text{ m/s}$
E_+	(1300, 100) V/m	minority	mass 3, $Z = 1$, 50%
E_-	(3900, 1) V/m	majority	deuterium, 50%

2 The steady distribution

Figure 1 shows the converged distribution. The RF builds a pronounced perpendicular tail: f falls by some fourteen decades between the bulk and $v_\perp \simeq 25 \text{ Mm/s}$, and the contours are strongly elongated in $v_\perp$, the signature of a J_1 -weighted quasi-linear operator depositing its energy in a narrow band at high perpendicular velocity. The parallel direction is barely heated.

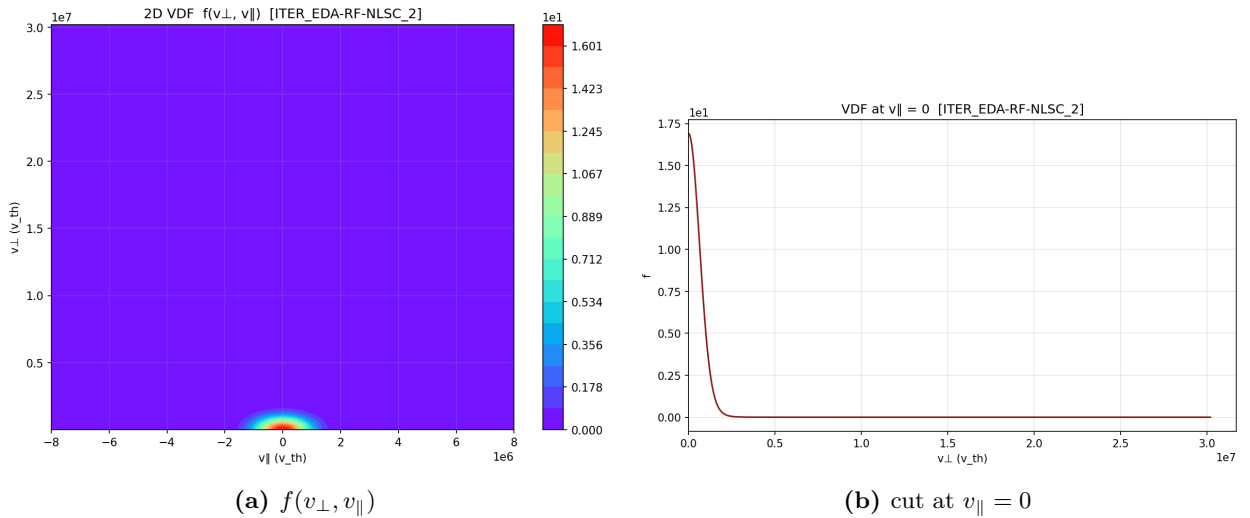


Figure 1: Converged minority distribution. The perpendicular tail extends to $\sim 25 \text{ Mm/s}$, some forty times the 0.62 Mm/s thermal velocity of the initial Stix Maxwellian.

3 Approach to steady state

The effective temperature rises from the 12.2 keV initial condition to 32.59 keV, and the anisotropy from 50.7% — essentially isotropic — to 73.9% (Fig. 2). Both saturate by $t \simeq 8$ s. The two quantities are not equivalent: T_{eff} nearly triples while the density-characteristic temperature T_n moves only from 12.1 to 13.7 keV, because T_n is set by the bulk of the population and T_{eff} by its second moment, which the tail dominates.

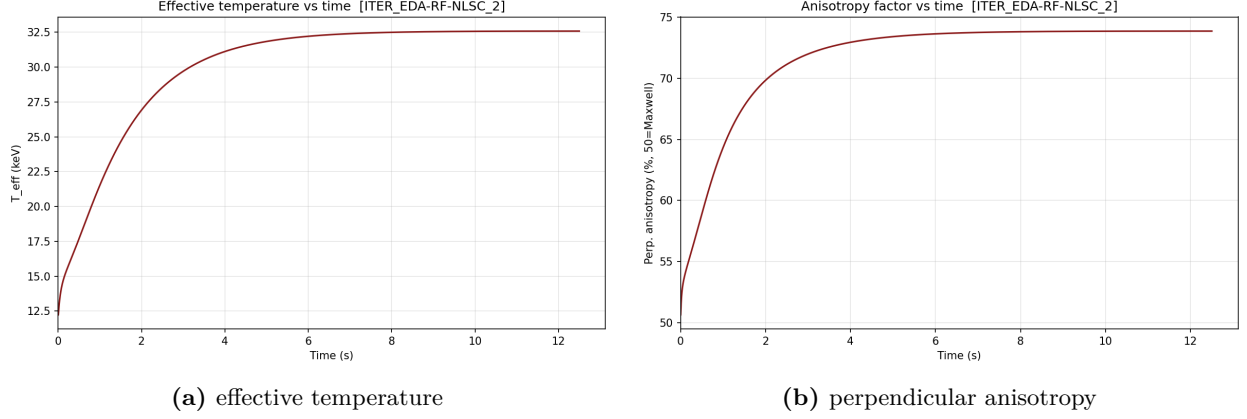


Figure 2: Approach to steady state; 50% anisotropy would be isotropic.

4 Power balance and timescales

At steady state the RF supplies 1.8166 MW/m^3 and the collisions remove 1.8166 MW/m^3 , closing to 0.003% (Fig. 3, left). The split is 1.075 MW/m^3 to the background ions and 0.742 to the electrons, or 59:41. That ordering is expected here: the critical energy for this minority is $E_c \simeq 106 \text{ keV}$, and at $T_{\text{eff}} = 32.6 \text{ keV}$ the tail sits well below it, so ion drag still leads. Self-collisions carry no net energy — they remove 0.176 MW/m^3 from the perpendicular direction and return exactly the same to the parallel, closing to 0.014% — but they are what couples the RF-driven perpendicular tail to the parallel direction and hence to the bulk.

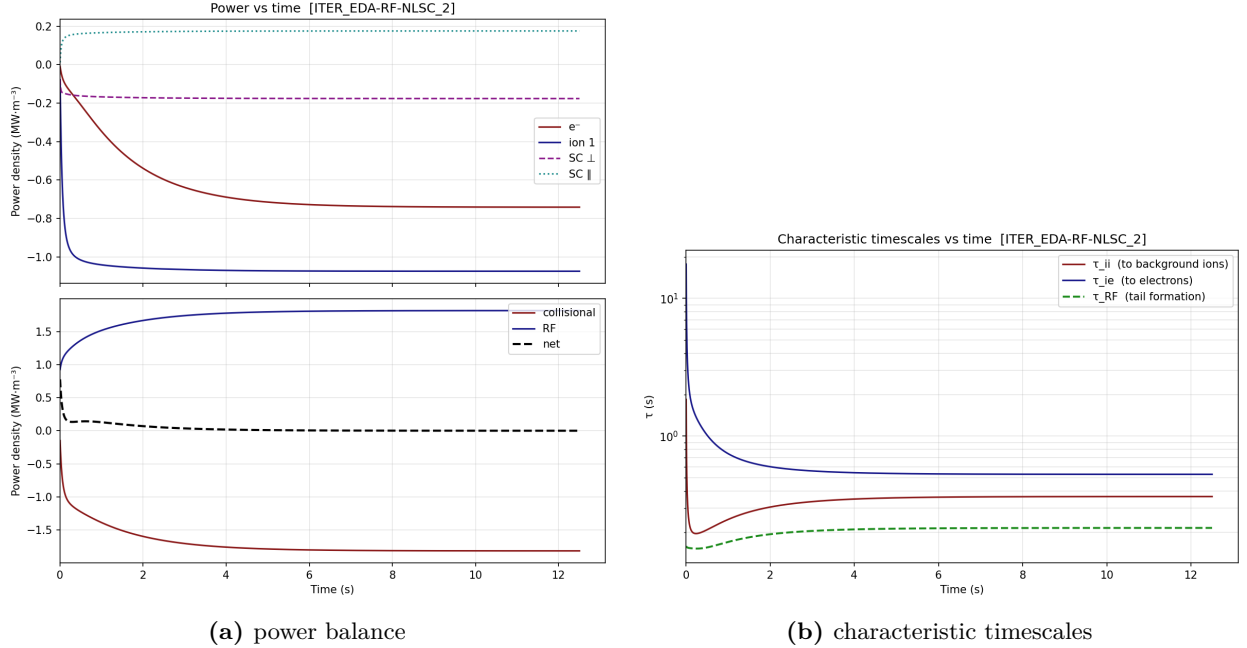


Figure 3: Power channels and the corresponding effective timescales $\tau = nT_{\text{eff}}/P$. The identity $1/\tau_{\text{RF}} = 1/\tau_{ii} + 1/\tau_{ie}$ holds to 0.003%, which is the power balance restated.

Read as timescales (Fig. 3, right), the RF fills the tail on $\tau_{\text{RF}} = 0.216 \text{ s}$ while the ions drain it on $\tau_{ii} = 0.364 \text{ s}$ and the electrons on $\tau_{ie} = 0.528 \text{ s}$. The RF is therefore the fastest process, by a factor 1.7 over the quicker of the two loss channels, which is why a substantial supra-thermal

population is sustained. All three fall steeply during tail formation, since a cold minority stores little energy relative to the power it can shed.

5 Summary and numerical quality

Table 2: Steady-state results.

T_{eff}	32.59 keV	P_{RF}	1.8166 MW/m ³
T_n	13.73 keV	P_{coll} (ions)	−1.0751 MW/m ³
anisotropy	73.86%	P_{coll} (elec.)	−0.7415 MW/m ³
τ_{RF}	0.2156 s	τ_{ii}	0.3643 s
τ_{ie}	0.5281 s	E_c	106 keV

The run stopped on the convergence criteria rather than exhausting its step budget, and the residual drift in T_{eff} over the final quarter of the run is below 0.3%. The minority density changes by +0.127% over the whole run: small, but not zero, and attributable to the slow particle gain of the non-linear self-collision operator, whose rate scales with the minority density. With the observed $d \ln T_{\text{eff}} / d \ln n \simeq -0.7$ this biases T_{eff} by under 0.1%. Wall truncation of f is 4.9×10^{-11} of the peak, so the velocity domain is ample.

Case `nominal_#2-REF`; figures from `figs/`, generated by `fp2d_plot.py` from the traces in `results/`.